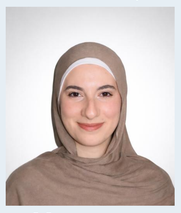
Accuracy of Single-Stage Digital Denture Workflow: An Arch-Specific Comparative Study

FIRST TIME PRESENTER

Deema Suleiman, DDS Candidate

Department of Comprehensive Dentistry, Louisiana State University School of Dentistry, New Orleans, LA, USA

LSU Health NEW ORLEANS School of Dentistry



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Introduction

Complete denture fabrication conventionally requires multiple appointments: a preliminary alginate impression, custom tray fabrication, a final elastomeric impression and a baseplate fabrication, creating a burden for elderly and medically compromised patients.

At LSU School of Dentistry, the preliminary alginate cast is scanned to design a 3D-printed custom tray. This raised the question: is that scan accurate enough to serve as the design reference for the denture baseplate, eliminating the final impression appointment entirely?

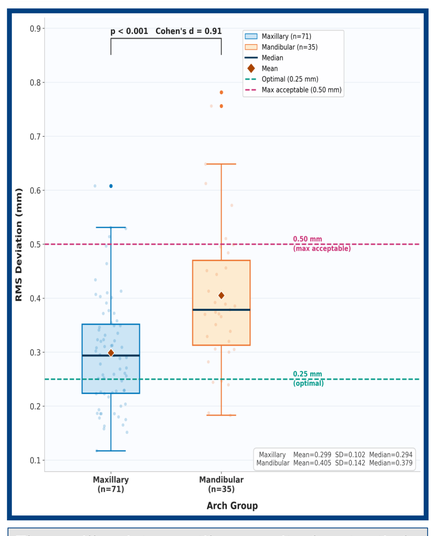
Published research places clinically acceptable accuracy window for complete denture fabrication at approximately 0.25-0.5 mm.[1]

Methods

1) Alginate impression in metal perforated stock trays poured in Type III dental stone within 20 minutes and faculty approved.
2) Custom tray designed in exocad from alginate cast scan. 3D printed with a 1.5 mm uniform spacer. Tray extends to full vestibular depth, so no border molding required.
3) Light body polysulfide final impression with mucostatic technique and poured in Type III dental stone.
4) 3D superimposition: both casts scanned and co-trimmed in Meshmixer to ensure identical surface regions compared.
5) Deviation analysis: ICP best-fit alignment in CloudCompare.

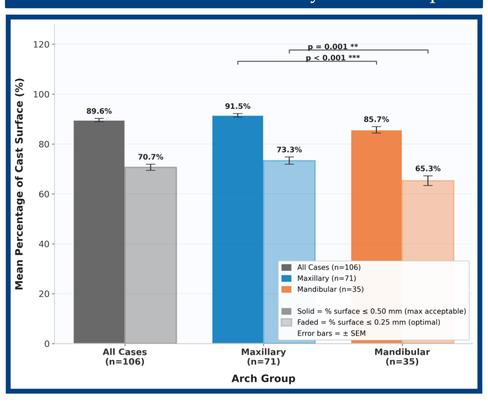
Results

Figure 1: RMS Deviation by Arch Group: Preliminary Alginate Cast vs. Final Polysulfide Cast



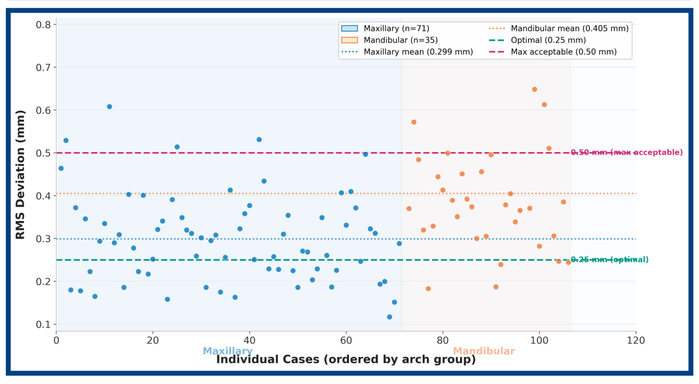
The maxillary interquartile range sits almost entirely below the 0.50 mm maximum acceptable clinical threshold, indicating consistent and predictable accuracy across cases. The mandibular box shows substantially greater spread, with outliers approaching 0.80 mm, nearly double the maximum acceptable limit, reflecting the unpredictable localized surface displacement that characterizes lower arch cases.

Figure 2: Percentage of Cast Surface Within Clinical tolerance Thresholds by Arch Group



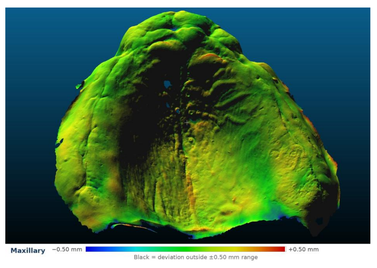
Maxillary cases significantly outperformed mandibular cases at both the 0.50 mm maximum acceptable threshold (91.5% vs. 85.7%; $p < 0.001$) and the 0.25 mm optimal threshold (73.3% vs. 65.3%; $p = 0.001$). The gap between arch groups is the direct consequence of the greater surface variability observed in mandibular cases. Error bars represent ± SEM.

Figure 3: Individual Case RMS Deviation :All 106 Cases



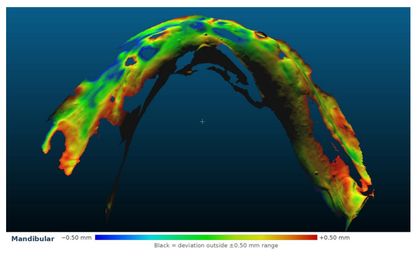
Each point represents a single arch. Maxillary cases (blue, n=71) cluster consistently below both clinical thresholds. Mandibular cases (orange, n=35) show greater scatter above the thresholds, with two outliers near 0.80 mm illustrating the extreme end of mandibular variability.

Figure 4: Maxillary Deviation Map: Representative Case (CloudCompare, ±0.50 mm Scale)



Three-dimensional superimposition of a preliminary alginate cast overlaid on its corresponding final polysulfide cast. The color scale spans ±0.50 mm; black regions represent areas where deviation exceeded the maximum acceptable threshold. The predominantly green-yellow distribution across the denture-bearing surface reflects moderate, relatively consistent deviation within the acceptable range.

Figure 5: Mandibular Deviation Map: Representative Case (CloudCompare, ±0.50 mm Scale)



Three-dimensional superimposition rendered on the same fixed ±0.50 mm color scale. The mixed red, orange, and green tones are distributed unpredictably across the ridge and vestibular surfaces, with areas of close fit sitting immediately adjacent to areas of high deviation. The extensive black regions indicate that a substantial portion of the cast surface exceeds the maximum acceptable threshold, visually illustrating the localized, inconsistent displacement captured statistically by the SD metric.

Conclusions

Maxillary: single-stage workflow is supported. Mean RMS 0.299 mm and 91.5% surface coverage within 0.5 mm (within clinically accepted accuracy window). Alginate cast scan already taken for custom tray design is sufficiently accurate to also serve as the baseplate design reference, potentially eliminating the final impression appointment for maxillary cases.

Mandibular: retain two-stage protocol. Greater RMS, higher SD, lower tolerance percentages and outliers; all five outcome measures consistently indicate that mandibular alginate impression carry unacceptable unpredictability for single-stage fabrication.

Unpredictability is the problem: both arches deviate by a similar average amount. The mandibular problem is that deviation is spatially inconsistent, some spots nearly perfect while other substantially off, making it impossible to predict where the fit will fail before the patient is in the chair.

Patient benefit potential. Eliminating the final impression appointment for maxillary cases reduces visits, chair time, material costs, and burden, which is particularly meaningful for elderly or medically compromised patients

Literature cited

1. Hayakawa I, et al. Digital assessment of preliminary impression accuracy for edentulous jaws. *J Prosthet Dent.* 2016;115(3):368-374.

Acknowledgments

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Further information

Software:

CloudCompare v2.13.2
Kharkiv · Meshmixer v3.5.474 · PrimeScan (Dentsply Sirona) · exocad · SprintRay Pro · Microsoft Excel 2024